

□ CHAPTER 3: Project Planning & Epic Framework

3.1 Systematic Project Lifecycles

[Epic 1: Architecture] —► [Epic 2: EDA Pass] —► [Epic 3: Pre-Processing] —► [Epic 4: Model Building] —► [Epic 5: Cloud Deployment]

3.2 Formal Epic Breakdown Tracking

- • Epic 1: Data Collection & Architecture Design — Source, download, and configure target risk qualification data files in local staging environments.
- • Epic 2: Visualizing & Analyzing Data (EDA) — Implement Pandas profiling modules to evaluate missing entries and structural data typing anomalies.
- • Epic 3: High-Fidelity Data Pre-Processing — Execute null-value imputation blocks and implement minority-class oversampling to prevent algorithmic optimization bias.
- • Epic 4: Model Building & Comparative Benchmarking — Perform training runs across Decision Tree, Random Forest, KNN, and XGBoost structures.
- • Epic 5: Production Application Implementation — Design responsive user form assets, combine features with a localized Flask container router, and package internal configurations via Docker files.